

Program: Diploma in Automation and Robotics	
Course Code: 4331	Course Title: Mechanics of Robotics
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To introduce the functional elements of Robotics.
- To understand the basic knowledge about kinematics of machines.
- To study basic mechanical drives for power transmission.
- To familiarize the knowledge of direct and inverse kinematics.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic principles and theorems of Engineering Physics		Physics	1 & 2
Forces and its applications		Engineering mechanics	2
Basic machine elements design features		Mechanics of materials	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic knowledge of kinematics of machines	10	Applying
CO2	Discuss the different mechanical drives used for power transmission	13	Applying
CO3	Familiarize with anatomy, specifications and applications of Robots	10	Understanding
CO4	To impart knowledge on the direct and inverse kinematics	10	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2	2					
CO3	2	2					
CO4		2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic knowledge of kinematics of machines		
M1.01	Understand the basics of mechanisms of machine elements	4	Understanding
M1.02	Understand the concept of cam and follower	3	Understanding
M1.03	Discuss the basics of keys, couplings and shafts	3	Applying
Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed)			
CO2	Discuss the different mechanical drives used for Power transmission		
M2.01	Discuss different belt drives for power transmission	3	Understanding
M2.02	Understand the advantages and applications of different rope and chain drives	2	Understanding
M2.03	Impart knowledge about the gears and gear trains.	4	Applying
M2.04	Understand the basic concept of gyroscopic forces and mechanical vibrations	2	Understanding

M2.05	Discuss the concepts of balancing of forces	2	Applying
Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives -velocity ratio -classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gear nomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system -Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts			
CO3	Familiarize with anatomy, specifications and applications of Robots		
M3.01	Explain different components of robot	4	Understanding
M3.02	Choose appropriate Robotic configuration and gripper for a particular application	4	Understanding
M3.03	Familiar with various design considerations for the selection of a robot or an application	3	Understanding
Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control.			
CO4	To impart knowledge on the direct and inverse kinematics		
M4.01	Understand and apply the concept of forward and inverse kinematics of manipulator.	5	Understanding
M4.02	Students can able to apply fundamentals of mechanism for the design of new mechanisms	5	Applying
	Series Test – II	1	
Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse			

kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist.

Text / Reference:

T/R	Book Title/Author
T1	S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4 th edition
T2	Sadhu Singh – ‘Theory of Machines’ – Pearson Education – 2011 – 3rd Edition
R1	S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition
R2	John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005.
R3	Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis’, Oxford University Press, Sixth impression, 2010.